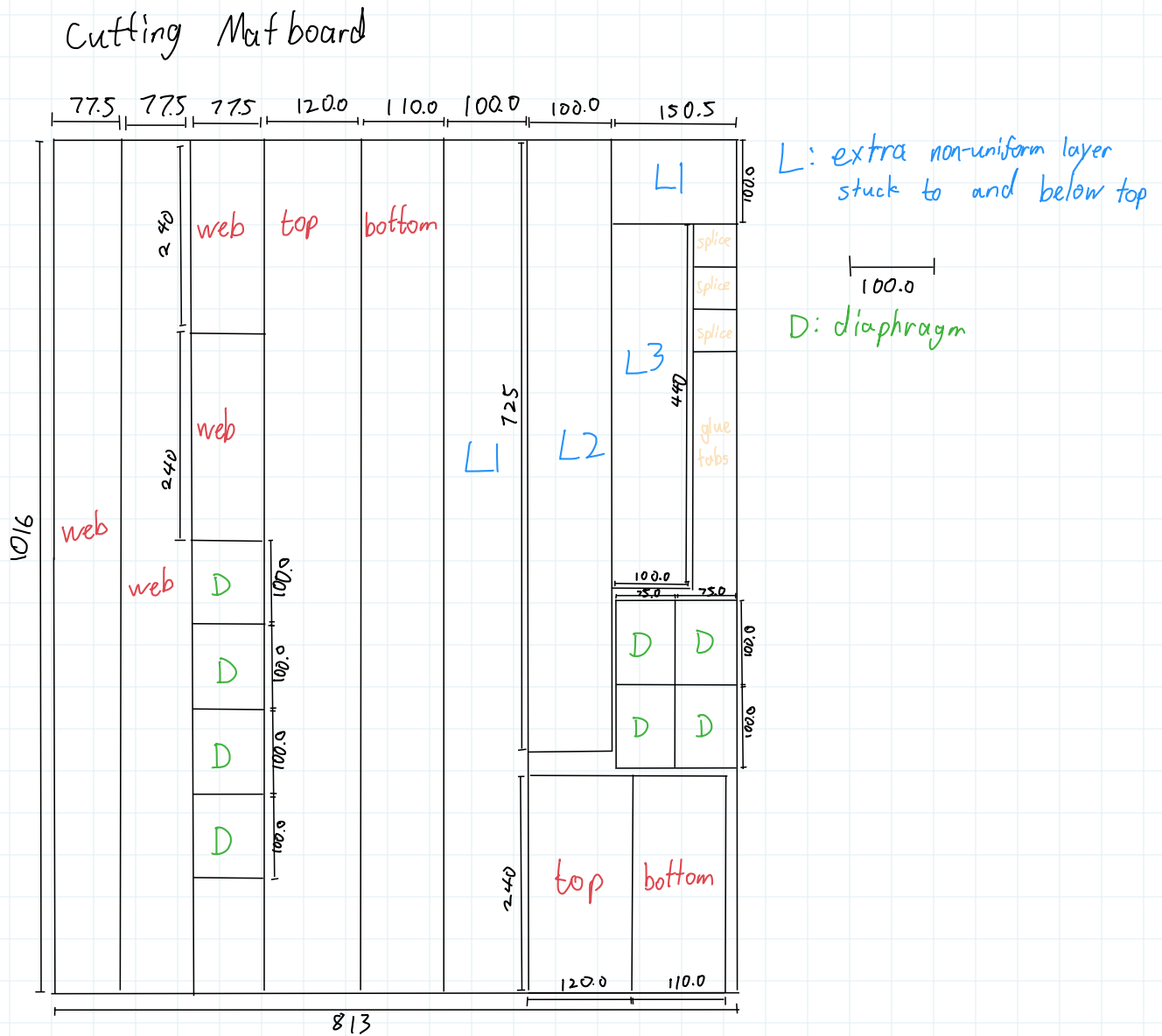




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Cross Section

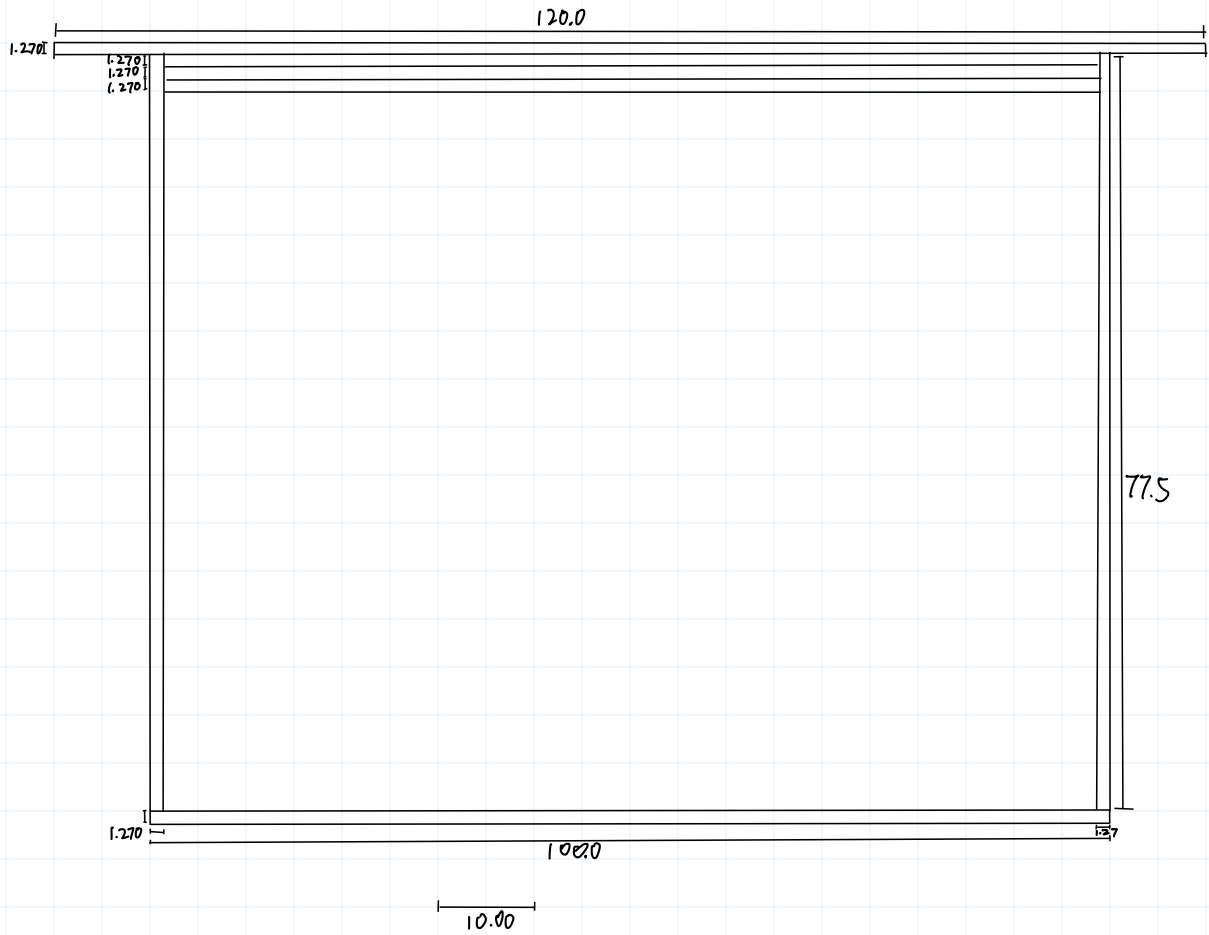




Diagram of a beam cross-section with dimensions:

- Total width: 1016
- Total height: 1000
- Top flange width: 250
- Top flange thickness: 50.0
- Web width: 50.0



1

NAME

COURSE NAME

2

CIV102 - Team 107 Construction Log

Further details may be found within the Design Report Section III

Construction Process

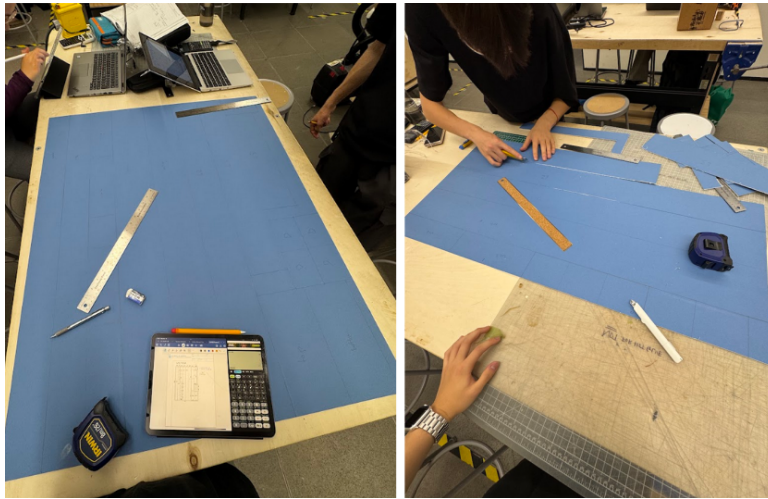
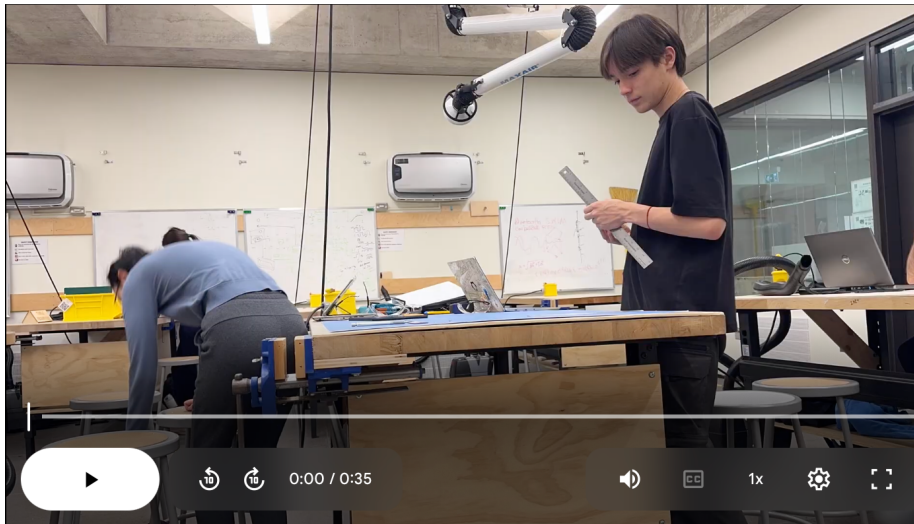


FIG. 1. Working on matboard tracing and cutting at the Myhal Centre Fabrication Facility with permission from the supervisors. As we understand, this does not qualify as a common space, and we made sure to obtain approval. Left: sketched lines and floorplan | Right: precision knife cutting process. 6:00 PM, Nov 19.

<https://drive.google.com/file/d/1n7kdDp04sYVEEbfvTPvGsN8ZWj4s4sa/view?usp=sharing>



Media 1: A video of the fast forwarded cutting process is included in the link above. 6:00 - 8:00 PM, Nov 19.

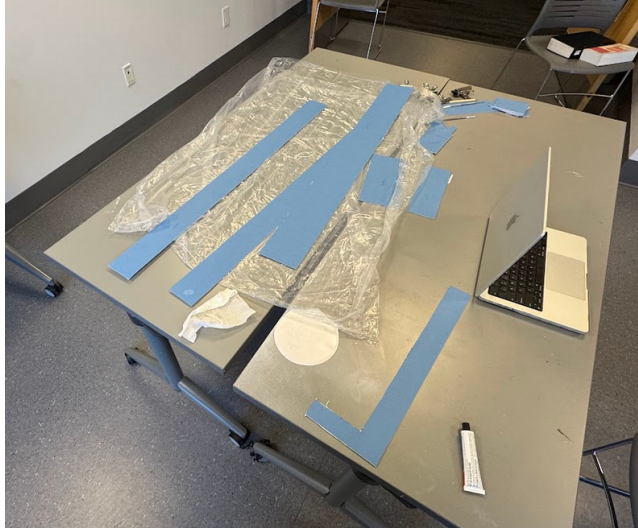


FIG. 2. Setting up for the day. Matboard pieces were placed on a tarp in the CampusOne crafts room. Gluing was done on a separate workbench. 3:15 PM, Nov 20.

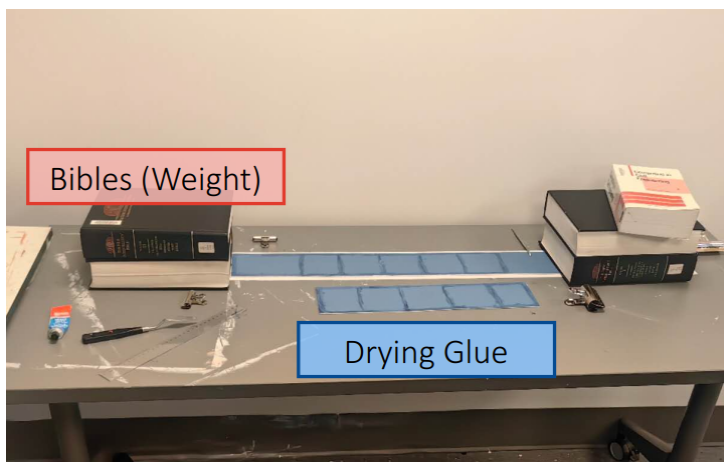


FIG. 3. Gluing the top track layers. As our design consisted of a 4 layer top, we glued them sequentially. 4:00 PM, Nov 20.



FIG. 4. Gluing the web. The web was glued following the tracks. 5:00 PM, Nov 20.

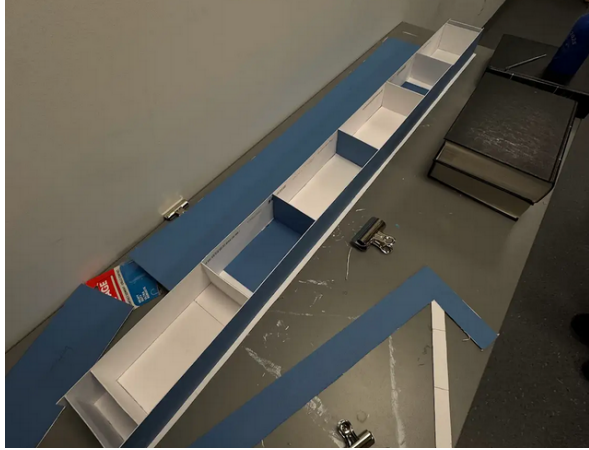


FIG. 5. Gluing the diaphragms. Following the web, diaphragms were glued into place. 6:00 PM, Nov 20.



FIG. 6. Pressure being applied following the application of the soffit. After attaching the diaphragms, we attached the bottom piece, then applied strong pressure along the length of the bridge. 8:00 PM, Nov 20.

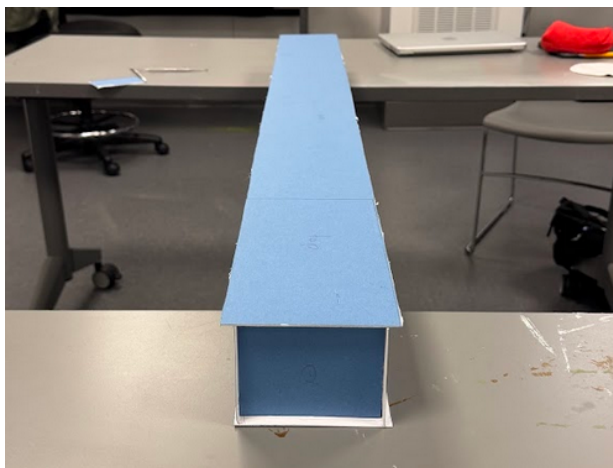


FIG. 7. Bridge as of the end of day on Nov 20th. 9:00 PM, Nov 20.

Contact Cement Safety

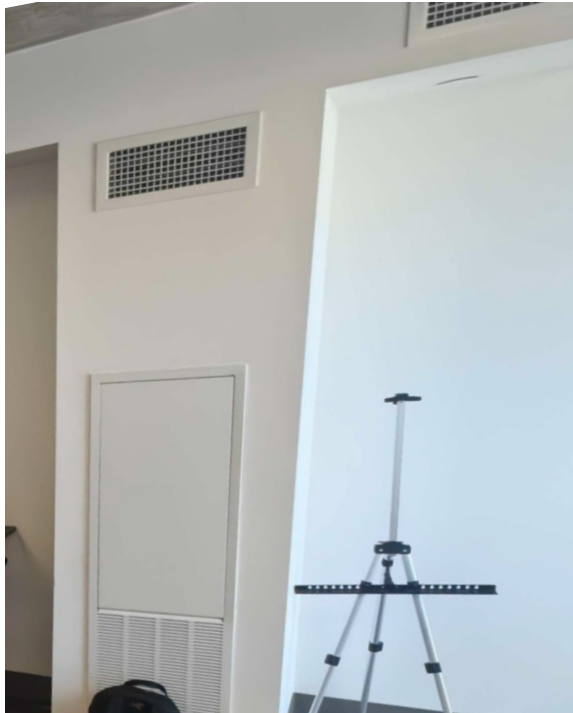


Figure 8: HVAC System in Room Provided Adequate Ventilation. The HVAC noise was noticeable implying strong circulation. We waited outside while the contact cement was drying to minimize exposure. Nov 20, 2025 (Full day)

Working Environment Cleanliness

We worked in a dedicated crafts room on the dirtiest table available. We avoided using a tarp as we found it shifted under the matboard. This was approved by CampusOne admin explicitly.

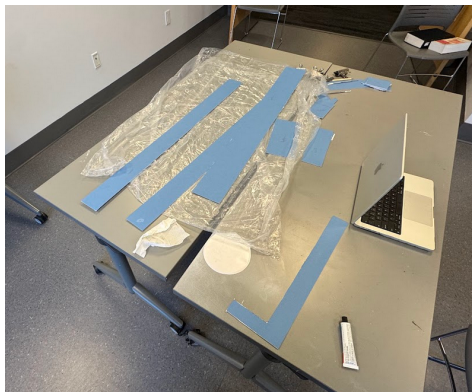


Figure 9: Keeping the main table clean. We placed a tarp here even though these pieces of matboard were not dirty.

Request for excuse: while we were very clean, and actually made sure the room was in the original state before leaving (spent roughly 10 minutes returning items, moving tables, cleaning scraps, etc.), we unfortunately did not take before-after photos. Please trust us that we cleaned the room very well.

As of Nov 23rd, 9 PM, there is unexpectedly another CIV102 group occupying the room making it hard to obtain photos before the assignment is due.

Additionally, the room is a dedicated arts and crafts space meant for private residents (which we are), and was already significantly dirty (paint stains may be seen in numerous figures). There is no need to worry that we dirtied (which we didn't!) a study space.

Given the above, we respectfully ask to be excused from the before-after photo sections of this assignment!

Best,
Luyu, Alan, Henry

Team 107 Time Log

Tasks	Date	Alan	Luyu	Henry	Total
Report					
Outline and Introduction	Nov 17	1	1	2	4
Design Iterations #1	Nov 18	0	2	3	5
Design Iterations #2	Nov 21	5	2	1	8
Construction Process	Nov 21	0	3	2	5
Introduction/Conclusion	Nov 21	0	2	1.5	3.5
Formatting to Latex/Editing	Nov 22	0	2	3	5
Final Editing	Nov 23	1	2	5	8
Drawing					
Making Figures/Plots #1	Nov 18	1	2	1	4
Making Figures/Plots #2	Nov 19	2	1	0	3
Construction Plans	Nov 19	2	0	0	2
Bridge Diagram #1	Nov 19	2	0	0	2
Bridge Diagram #2	Nov 21	1	0	0	1
Calculations					
Creating BME and SFE	Nov 12	2	0	0.5	2.5
Rough Estimations	Nov 17	2	2	0	4
Verifying design iterations / optimization #1	Nov 18	3	2	0	5
Verifying design iterations / optimization #2	Nov 19	5	0.5	0.5	6
FOS Calculations Code	Nov 20	3	0	0	3

Required Hand Calculations #1	Nov 20	1	0	0	1
Required Hand Calculations #2	Nov 21	2	0	0	2
Formatting Calculations/Code	Nov 21	1	0	0	1
Verifying All Calculations	Nov 23	1	2	2	5
Construction					
Cutting	Nov 19	0.5	3	3	6.5
Gluing #1	Nov 20	0	8	7	15
Gluing #2	Nov 22	0	2	0	2
Gathering Equipment	Nov 20	1	1	1	3
Others					
Deliverable #1	Nov 10	1	1	1	3
Reviewing and feedback	Nov 17	0.5	1	1	2.5
Total (hrs)		38	37.5	34.5	

Signature 1: Alan Signature 2: Luyu Signature 3: Henry Kim